Event-Based Trading: *Building Superior Trading Strategies with State-of-the-Art Information Extraction Tools*

Complete Research Paper

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Abstract

Advanced sentiment analysis systems, which typically produce a positive or negative signal on firms for financial traders, consider financial events when calculating aggregate sentiment scores. However, these systems typically produce overly noisy signals for traders because they aggregate the information culled from the events rather than address each distinct event-type differently (all event-types receive the same weight in a text's final sentiment score). We demonstrate that event-types have different informative values for traders, and specific event-types may be particularly relevant for specific investment periods. In this work we evaluate and compare the effects of event-type signals and sentiment analysis signals, automatically extracted from a given text, on stock prices in different investment periods. We show that traders using trading strategies based on event-types and sentiment analysis obtain superior performance compared to acceptable benchmarks.

Introduction

Current technological advancements in data science open new avenues to financial traders, who are constantly looking for new strategies, methods and tools to assist them in making better investment decisions. Powerful computers regularly monitor and analyze structured data to provide signals to traders, and High Frequency Trading algorithms even execute trades on their behalf, with the goal of generating arbitrage profits from short-term fluctuations (Terrence and Riordan 2013; Treleaven et al. 2013). Although traders commonly use computers to process and analyze financial structured data, they less frequently rely on computational abilities to process and analyze unstructured data. Technological advancements in the field of Natural Language Processing (NLP) offer traders with tools necessary to analyze unstructured financial data. Sentiment analysis, for example, which has captured the attention of financial traders around the world, offers great value in determining the tone of large amounts of textual data about firms, markets, commodities, and other financial instruments (Feldman 2013; Loughran and Mcdonald 2016). By providing additional signals to traders that can be considered when devising their trading strategies, sentiment analysis can beneficially support trading decisions. Most existing studies on sentiment analysis in the field of finance look at the aggregate sentiment score assigned to firms after analyzing texts from trade or social digital media (Bollen et al. 2011; Feldman et al. 2010; Tetlock et al. 2008). Irrespective of the accuracy of the applied sentiment analysis method, most systems tend to miss one crucial factor in the analysis, which is context. By counting positive and negative words, or even by performing a flawless sentiment analysis, one can at most obtain the overall tone of the article, which, we argue here, can be a very noisy signal to traders. In this work, we suggest that one should not treat all sentiment expressions evenly, and different event-types that embed sentiments should be considered and used as trading signals relevant for specific investment periods.

News of financial events is presumed to affect stock prices. There are different opinions in the literature on the extent to which news is reflected in stock prices and the time lag involved before such news is reflected. These questions are fundamentally tied to Eugene Fama's Efficient Market Hypothesis (EMH) (Fama 1970). Many studies have since attempted to identify methods in which publicly available information can, nevertheless, be used to obtain excess returns (Antweiler and Frank 2004; Davis et al. 2012; Finnerty 1976; Grossman and Stiglitz 1980). These studies argue against Fama's hypothesis that states that it is impossible to beat the market because prices already incorporate and reflect all relevant information.

In this work, we analyze which events reported on digital media can be valuable signals for traders in making trading decisions and whether they enhance or diminish the value of sentiment analysis signals. Subsequent questions are whether events should be considered separately or aggregately, and for which periods relevant securities should be held after obtaining such signals. To answer these questions, we used VIP (Visual Information Extraction Platform), an automatic information extraction platform, for the text analysis and the Quantopian Research Platform for the performance analysis. VIP processes, analyzes, and converts unstructured financial data into structured data sets of stock-specific sentiments tied to business events. This tool allows us to identify various sentiments and events that are related to public companies. Using Quantopian, one can create trading algorithms and test them against historical data (a process known as backtesting). We also used Alphalens, a component of the Quantopian Research Platform, which was designed to analyze alpha factors.

Related work

The question of how to exploit unstructured firm-related digital data has occupied many financial economists and computer scientists. Early work argues that by merely observing the tone of the text about a particular firm on digital media, traders can obtain a valuable signal. For example, Tetlock et al. (2008) found that the occurrence of negative words in news articles about firms could function as a valuable signal for traders. With advances in the field of NLP, researchers have adopted more sophisticated approaches to sentiment analysis and information extraction, and applied them on financial texts (Davis et al. 2012; Heston and Sinha 2015; Li et al. 2014; Schumaker and Chen 2009; Zhang and Skiena 2010). Boudoukh et al. (2016) showed that public firm-level news is a meaningful component of stock return variance. Distinguishing between information that is relevant to the firm’s fundamentals (e.g., analyst recommendations, acquisitions, products, etc.) and irrelevant information (e.g. executive changes, dividends, etc.), they argued that only certain events should be considered relevant signals. This level of granularity is, however, inadequate on two counts. First, after distinguishing between events that are likely to impact stock prices and those that are not, these researchers treated all relevant events equally in their analysis. Rather than considering the individual impact of every event - type, the researchers looked at the aggregate effect of all relevant events. Second, we argue that events should be further classified to distinguish between subsets of events. For example, product events might involve product trials or product recalls. Each event-type would obviously have a very different impact on stock prices. Boudoukh et al. (2016) do not make such distinction. Ding et al. (2014) used a deep neural network model in order to predict stock price movements, but similarly to Boudoukh et al. (2016), they aggregated events' sentiments to determine the polarity of the news.

To the best of our knowledge, no other study considers the impact of individual events, identified by an automatic Information extraction platform, and compares the excess return generated when such signals are used separately or together with the sentiment scores (tones), over various time horizons.

Data and data processing

To test which event-types on digital media might be valuable signals for investors making trading decisions, whether events should be used as isolated or aggregate signals, and to compare event signals to sentiment signals we obtained three corpora:

- *Corpus A*: 6,750 articles[1] on various firms and indices
- *Corpus B*: 1,000,000 articles [2]on various firms
- *Corpus C*: 1,567,260 Dow Jones newswires articles[3].

We processed these corpora using VIP to extract sentiments and events related to different firms. VIP processes, analyzes, and converts unstructured financial data into structured data sets of stock-specific sentiments tied to business events. VIP uses state-of-the-art NLP

[1] downloaded from Yahoo Finance (https://finance.yahoo.com) and Google Finance (https://www.google.com/finance)
[2] downloaded from Yahoo Finance (https://finance.yahoo.com) and Google Finance (https://www.google.com/finance)
[3] Downloaded from the Dow Jones newswires website (http://www.dowjones.com/products/dj-news). The articles were published between March 2011 and March 2016 and discuss 613 different firms traded on the NYSE, NASDAQ and AMEX that are mostly included in the S&P 500 index.

techniques, specifically a proprietary, leading-edge pattern language designed to identify relevant linguistic patterns in context-free grammar involving sentiments or business events. The core of the VIP system is an integrated sentiment analysis engine that incorporates knowledge from multiple sources, and performs a relevance analysis to determine sentiment and event scores of financial content of potential significance for financial traders. Using VIP, we generated the following outputs for every company: *Daily sentiment score* – ranges between -1 and 1 and is calculated as: $Score = \frac{P-N}{P+N+3}$ (where P and N represent the number of positive and negative sentiment expressions, respectively); *Time-weighted sentiment score* – the sum of all sentiment scores of a particular ticker, while weighting older articles with a decay factor of a half-life of 10 days.; and *Daily event count* – number of positive and negative mentions of events of a particular type on a certain traded company, e.g., number of positive or negative Financial Results mentioned on a given company in a given day.

To conduct a financial analysis of VIP's output, we followed a few preprocessing steps: we removed tickers that did not exist on the first date of the analysis or that since changed. (this step is required for mapping symbols to security objects in Quantopian); we calculated the daily event score (positives count minus negatives count) for each event-type; we calculated the aggregated daily event score for all event-types, for each ticker; we normalized the event score and the aggregated daily event scores by log(n+1) to handle skewness of the data. We ended up with the following daily signals for each company in our corpora: *Daily Sentiment Score*; *Time-weighted Sentiment Score*; *Normalized Aggregated Event Score*; and *Normalized Event Score by event-type*.

The Quantopian Research Platform

The Quantopian Research platform[4] is an IPython notebook environment that is used for research and data analysis during algorithm creation. It is also used for analyzing the past performance of algorithms. The IDE (interactive development environment) is used for writing algorithms and kicking off backtests using historical data. We uploaded the signals described in the previous section to the Quantopian Research platform and used the Alphalens Python package for the analysis. Alphalens[5] is a Python package for performance analysis of alpha factors, which can be used to create cross-sectional equity algorithms. Building a rigorous workflow with Alphalens enhances the robustness of backtesting strategies and reduces proneness to overfitting, both of which are important parameters in algorithm evaluation.

For each signal, we conducted the Alphalens analysis, as follows: Eliminate all nulls; Map symbols to security objects in Quantopian; Index the dataframe properly, such that it is similar to pipeline output; Convert the dates to UTC; Obtain pricing data from the earliest date up to a month after the final date; Use the signal of t-1 and trade on the open date of t; Create the factor tear sheet using the signal and six periods (1, 5, 10, 15, 20 and 30 days).

Experiments

To evaluate the economic value of events and sentiment signals generated from an automated information extraction platform, we conducted a set of experiments. First, we

[4] https://www.quantopian.com/research
[5] http://quantopian.github.io/alphalens/

tested whether the polarity of specific events can be used as a signal for financial traders. More specifically, we identified which positive (negative) events were followed by an increase (decrease) in stock prices and after what interval. Second, we looked at whether buying specific stocks when certain events appear in digital media outperforms buying a benchmark portfolio for a given period. Third, we tested whether event-types perform differently in different sectors. Finally, using the Quantopian Research Platform, we analyzed event-type signals extracted from a large scaled dataset, and determined the event-type signals with the potential to generate superior alphas for traders, and in which holding periods.

Experiment I – Event Polarity as a Signal

In the first experiment, we evaluated whether event polarity could be used as an indicator of future positive and negative returns, in different holding periods. For this experiment, we used Corpus A. We tested the correlation of the polarity of these events with future stock prices, that is, whether an event classified as positive (negative) was followed by an increase (decline) in stock price. We looked at the changes in stock prices in time lags of 1, 2, 5, 7, 10, 15, and 20 trading days. The results of this experiment for the 10 most frequent events in our corpus are presented in Figure 1.

The results of this experiment clearly show that events have different effects over different time periods. For example, the effect of negative financial results seems to be strongest after one trading day, and fades off over time, while positive financial results seem to have a weaker effect after one trading day, but this effect becomes stronger over time

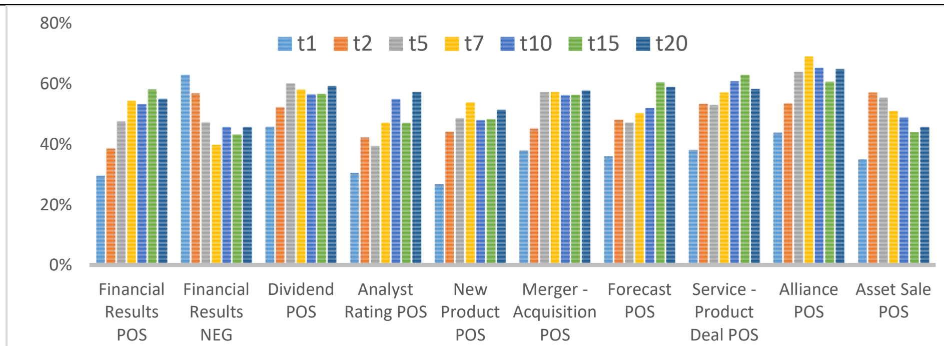

Figure 1: Predicting stock returns using event polarity, by event, by time period. The Figure shows the correlation between the events' polarity and the corresponding stock return after t-days.

Experiment II - Comparing performance to benchmarks

In the third experiment, we explored whether trading specific stocks when certain events appear in digital media outperforms trading a benchmark portfolio (Russel, 2000). We processed Corpus B with VIP , and simulated a situation in which traders buy stocks (sell short) when positive (negative) events occur, and hold them for different time periods.

We recorded cases when trading according to events significantly outperformed trading the benchmark portfolio. A Benjamini-Hochberg correction was applied to control for the fact that small p-values (less than 5%) may occur by chance, and lead to the incorrect rejection of true null hypotheses. The results are presented in Figure 3 below.

	t0	t1	t2	t3	t4	t5	t6	t7	t8	t9	t10	t11	t12	t13	t14	t15	t16	t17	t18	t19	t20	t21	t22	t23	t24	t25	t26	t27	t28	t29	t30
Negative Inside Sell Purchase (n=831,783)	+	+	+	+	+																										
Positive Alliance (n=821,837)																											+	+			
Positive New Product (n=814,790)		+																													
Positive Financial Results (n=786,557)		+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+						
Positive Merger and Acquisition (n=661,737)	+	+							+	+																					
Positive Contract - Agreement - Deal (n=579,513)	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Negative Financial Results (n=391,096)		+	+																												
Positive Forecast (n=390,819)		+	+	+	+		+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Positive Dividend (n=374,032)					+	+	+	+																							
Positive Facilities (n=371,548)	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Positive Investment (n=335,243)	+	+	+	+	+	+	+			+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+			
Positive Executive Change (n=334,256)	+	+																					+	+	+	+	+	+	+	+	+
Positive Asset Sale (n=292,888)			+	+											+	+															
Positive Analyst Rating (n=264,927)	+						+	+	+	+	+							+		+					+	+	+	+	+	+	+
Positive Inside Sell Purchase (n=255,826)					+	+		+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Positive Asset Acquisition (n=248,505)	+	+	+	+	+	+	+	+	+		+		+	+	+	+	+	+			+	+	+	+	+	+	+	+	+	+	+
Positive Service Product Deal (n=226,043)																															
Negative Forecast (n=226,016)		+	+																												
Negative Lawsuit (n=209,818)																													+	+	+
Positive Stock Buyback (n=172,491)																									+	+	+	+	+	+	+

Figure 3: Trading based on events compared to trading a benchmark portfolio. (+) signifies that trading on event signals significantly outperforms trading the benchmark portfolio.

Experiment III – Events in different sectors

In the fourth experiment, we tested whether similar event-types in different sectors have different economic value for traders. In particular we looked at the following nine sectors: *SnPEnergy*, *SnPTech*, *Industrials*, *Staples*, *SnPMaterials*, *SnPFinance*, *TelecomUtilities*, HealthCare, Discretionary, and a set comprising all nine sectors.

For this experiment, we processed Corpus B with VIP and classified the companies by sector. We looked at three short-term trading periods of 1, 2, and 5 days. The effects of similar event-type in different sectors showed no significant differences for a trading period of one day. The adjacency matrices in Figure 4 below outline the significant differences between similar events in different sectors for trading horizons of 2 and 5 days

T=2

	SnPEnergy	SnPTech	Industrials	Staples	SnPMaterials	SnPFinance	TelecomUtilities	HealthCare	Discretionary	All Sectors
SnPEnergy										
SnPTech				0.04						
Industrials										
Staples		0.04				0.05		0.04		
SnPMaterials										
SnPFinance				0.05						
TelecomUtilities										
HealthCare				0.04						
Discretionary										
All sectors										

T-5

	SnPEnergy	SnPTech	Industrials	Staples	SnPMaterials	SnPFinance	TelecomUtilities	HealthCare	Discretionary	All Sectors
SnPEnergy										
SnPTech					0.035					
Industrials										
Staples										
SnPMaterials		0.035				0.019		0.027	0.018	
SnPFinance					0.019					
TelecomUtilities										
HealthCare					0.027					
Discretionary					0.018					
All sectors										

Figure 4: Adjacency matrix of significant differences between similar events in different sectors (p-values)

The results indicate that for a time period of two days, similar event-type signals generated significantly different effects in Staples, SnPFinance, and SnPTech sectors. For a five-day period, event-type signals generated significantly different effects in SnPMaterials and Health Care as well as SnPFinance and SnPTech. In general, each event-type signal generated a similar effect in most sectors, which led us to believe that sectors should not play an important role when considering event signals in short-term trading periods.

Experiment IV – Developing superior trading strategies using event detection

For our final experiment, we used Alphalens and the Quantopian Research Platform to conduct a return analysis when investing according to combinations of sentiment signals and various event-type signals, which were identified with VIP after processing Corpus C. We first compared between the alphas and betas resulting from investing according to a Daily Sentiment Score signal, Time-weighted Sentiment Score; signal and the Aggregated Event Score signal in various investment periods ranging from one day to one month. The results are presented in Table 1 below. Cells highlighted in green represent excess return compared to the returns based on the Daily Sentiment Score signal, and cells highlighted in red represent inferior return compared to returns based on the Daily Sentiment Score signal.

Table 1: Alphas and betas resulting from investing according to the Daily Sentiment Score signal, Time-weighted Sentiment Score signal and Aggregated Event Score signal. Annual Alpha (Beta).

	1	**5**	**10**	**15**	**20**	**30**
Daily Sentiment Score signal	0.296 (-0.024)	0.065 (-0.02)	0.03 (-0.018)	0.019 (-0.015)	0.015 (-0.017)	0.011 (-0.025)
Time-weighted Sentiment Score	0.228 (-0.061)	0.056 (-0.049)	0.032 (-0.056)	0.027 (-0.06)	0.027 (-0.065)	0.023 (-0.077)
Aggregated Event Score	0.272 (0.001)	0.054 (-0.011)	0.019 (-0.005)	0.01 (-0.008)	0.01 (-0.013)	0.01 (-0.003)

Of these three signals, the best alpha for the short run (shorter than 10 days) was obtained using the Daily Sentiment Score signal, which outperformed the alphas of Time-weighted Sentiment Score and Aggregated Event Score signals. In the longer run (10 days and longer), however, the alpha of the Time-weighted Sentiment Score signal outperformed the alpha of Daily Sentiment Score and Aggregated Event Score signals.

Since specific event-types do not occur on most days, it is not practical to use these event-types as daily trading signals, unlike the Daily Sentiment Score, Time-weighted Sentiment score and Aggregated Event Score signals. Therefore, to test their economic trading value of the last three signals, we test whether returns increase or decline when specific event-type signals are added to the Daily Sentiment Score signal. The results for the Daily Sentiment Score signal together with the 25 most frequent event-types in our corpus are presented in Table 2 below. When the signal from a particular event-types improves the economic value of the Daily Sentiment Score signal, the corresponding cell is highlighted in green. When the signal reduces the economic value of the Daily Sentiment Score signal, the corresponding cell is highlighted in red.

Several insights emerge from this experiment: First, signals based on events mentioned in digital media can be used as valuable signals for investors when devising their trading strategy, and all event-based signals used in this study seem to generate a positive alpha. Second, we classify several event-types by their impact on trading returns: events that have a positive long-term effect (e.g., "Lawsuits" and "New Products"), events with a positive short-term effect (e.g., "Dividends"), events that seem to have a positive effect for all time periods examined in this study (e.g., "Financial Results" and "Forecast"), and events that do not seem to have a positive effect in any time period (e.g., "Asset Sales" and "Workforce Changes"). Third, of the event-type signals that we tested, the "Financial Results" signal seems to have the most positive impact on the Daily Sentiment Score signal. Fourth, the low betas of all tested event-types indicate that event-based trading generates less volatile results compared to a benchmark.

Table 2: Alphas and betas resulting from an investment strategy based on Daily Sentiment Score and the 25 most frequent event-types in our corpus – annual Alphas (Betas).

	1	5	10	15	20	30
Financial Results	0.447 (-0.018)	0.093 (-0.02)	0.044 (-0.018)	0.029 (-0.02)	0.029 (-0.026)	0.029 (-0.034)
Inside Sell Purchase	0.272 (-0.014)	0.062 (-0.012)	0.026 (-0.008)	0.016 (-0.004)	0.016 (-0.006)	0.016 (-0.012)
New Product	0.283 (-0.027)	0.063 (-0.022)	0.028 (-0.02)	0.019 (-0.018)	0.019 (-0.021)	0.019 (-0.028)
Alliance	0.279 (-0.025)	0.063 (-0.023)	0.029 (-0.021)	0.018 (-0.019)	0.018 (-0.021)	0.018 (-0.026)
Merger and Acquisition	0.287 (-0.025)	0.062 (-0.024)	0.028 (-0.021)	0.017 (-0.018)	0.017 (-0.019)	0.017 (-0.024)
Forecast	0.435 (-0.016)	0.083 (-0.013)	0.036 (-0.001)	0.022 (-0.007)	0.022 (-0.003)	0.022 (-0.012)
Contract - Agreement - Deal	0.285 (-0.022)	0.06 (-0.022)	0.027 (-0.021)	0.018 (-0.017)	0.018 (-0.018)	0.018 (-0.024)
Analyst Rating	0.314 (-0.025)	0.069 (-0.021)	0.031 (-0.017)	0.021 (-0.016)	0.021 (-0.02)	0.021 (-0.028)
Executive Change	0.286 (-0.021)	0.061 (-0.017)	0.027 (-0.014)	0.016 (-0.012)	0.016 (-0.014)	0.016 (-0.021)
Dividend	0.3 (-0.025)	0.07 (-0.022)	0.028 (-0.022)	0.017 (-0.02)	0.013 (-0.022)	0.009 (-0.028)
Facilities	0.282 (-0.027)	0.062 (-0.022)	0.027 (-0.02)	0.018 (-0.018)	0.018 (-0.02)	0.018 (-0.028)
Investment	0.279 (0.023)	0.062 (-0.02)	0.027 (-0.017)	0.017 (-0.016)	0.017 (-0.018)	0.017 (-0.025)
Asset Sale	0.278 (-0.02)	0.06 (-0.018)	0.026 (-0.016)	0.016 (-0.011)	0.012 (-0.011)	0.008 (-0.02)
Asset Acquisition	0.29 (-0.022)	0.064 (-0.017)	0.028 (-0.015)	0.018 (-0.013)	0.014 (-0.015)	0.01 (-0.023)
Lawsuit	0.29 (-0.024)	0.064 (-0.018)	0.029 (-0.018)	0.019 (-0.016)	0.019 (-0.017)	0.019 (-0.024)
Service Product Deal	0.295 (-0.022)	0.064 (-0.019)	0.028 (-0.017)	0.019 (-0.014)	0.015 (-0.015)	0.01 (-0.023)
Stock Buyback	0.301 (-0.024)	0.068 (-0.02)	0.032 (-0.019)	0.021 (-0.016)	0.016 (-0.019)	0.012 (-0.026)
Senior Executive Change	0.291 (-0.023)	0.064 (-0.018)	0.029 (-0.017)	0.019 (-0.016)	0.015 (-0.019)	0.011 (-0.02)
Credit Debt Rating	0.295 (-0.025)	0.066 (-0.021)	0.03 (-0.019)	0.019 (-0.016)	0.016 (-0.019)	0.011 (-0.026)
Board Change	0.293 (-0.024)	0.065 (-0.021)	0.03 (-0.02)	0.019 (-0.015)	0.015 (-0.017)	0.011 (-0.024)
Price Target	0.312 (-0.025)	0.068 (-0.021)	0.031 (-0.019)	0.02 (-0.015)	0.017 (-0.018)	0.012 (-0.026)
Workforce Change	0.296 (-0.022)	0.065 (-0.018)	0.03 (-0.017)	0.019 (-0.014)	0.015 (-0.016)	0.011 (-0.024)
Settlement	0.293 (-0.021)	0.063 (-0.017)	0.028 (-0.017)	0.018 (-0.013)	0.014 (-0.015)	0.01 (-0.021)
Investigation	0.297 (-0.023)	0.065 (-0.018)	0.029 (-0.016)	0.019 (-0.016)	0.015 (-0.019)	0.011 (-0.026)
Product Update	0.295 (-0.025)	0.065 (-0.021)	0.03 (-0.019)	0.019 (-0.016)	0.016 (-0.017)	0.011 (-0.026)

Conclusions and future work

In this paper, we evaluated whether financial events extracted from digital media using an automatic information extraction platform constitute valuable trading signals. The effect of different events was determined in different time lags. We established that the polarity of different event-types is, indeed, reflected in the stock price in different time periods. We concluded that in contrast to aggregate polarity of events, which is typically used in sentiment analysis systems, event-types should be used as isolated signals. We further used a benchmark portfolio and demonstrated that superior returns can be obtained when considering different event-types for different investment periods. We found that both sentiments and events can generate positive alphas. We further found that Daily Sentiment Score signal is a better signal than Time-weighted Sentiment Score signal in the short term, and vice versa in the long term. When used aggregately, event-related information produces a noisy signal that does not improve the alpha of the Daily Sentiment Score signal. Using specific event-types, however, enhances the results of a strategy based on the Daily Sentiment Score signal. Event-types impact the Daily Sentiment Score signal differently in different time periods. We also compared the effect of the same event-type in different sectors, and in most cases, we found that the same event-types had a similar impact in all sectors. Consequently, we believe that sector-based information is less important in developing event-based trading strategies.

Future work might further investigate how to use automated information extraction platforms to further enhance event-based trading strategies, or might focus on determining the effect of events in non-financial domains.

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